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Unveiling the American Auto Landscape

Introduction

In this project we will see the world of US car sales, offering an overall look at how vehicles move across the country. By using various data in our dataset we'll find trends and patterns that will show consumer preferences, regional variations, and the overall health of the US auto industry. We are here to find out who dominated the car market in the US.

Dataset

Car id- Specific unique identification number for a car.

Date- The day on which the specific car was purchased.

Customer Name- Name of the person who purchased the car.

Gender- The gender of the buyer.

Annual Income- The income of the buyer.

Annual Expenditure- The expenditure of the buyer.

Dealer Name- Name of the dealership who sold the car to the customer.

Company- The name of the brand the customer brought.

Model- The specific model of the brand purchased by the buyer.

Colour- The colour of the car sold to the buyer.

Price- The price at which the dealer buys the cars from the manufacturer.

Selling Price- The price at which the car is sold to the customer.

Second hand price- Price at which the same model of the car can be brought (5 year later)

Dealer No- The unique id number of the dealer.

Body Style- The type of car being sold.

Phone- Phone number of the Buyer.

Location- Place where the car is being sold.

Calculated Fields

Count of model= No of units of that specific model sold

Count of company= No of units of that specific company sold

Sales= Count of model*Selling price of the model

Depreciation= Selling Price-Second hand price

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Depreciation %= (Depreciation/Selling Price) *100

Profit=Selling Price- Price

Profit Margin= (Selling Price-Price/Price) *100

Savings= Annual income- Annual Expenditure

Pre-processing Dataset

1) Correlation matrix

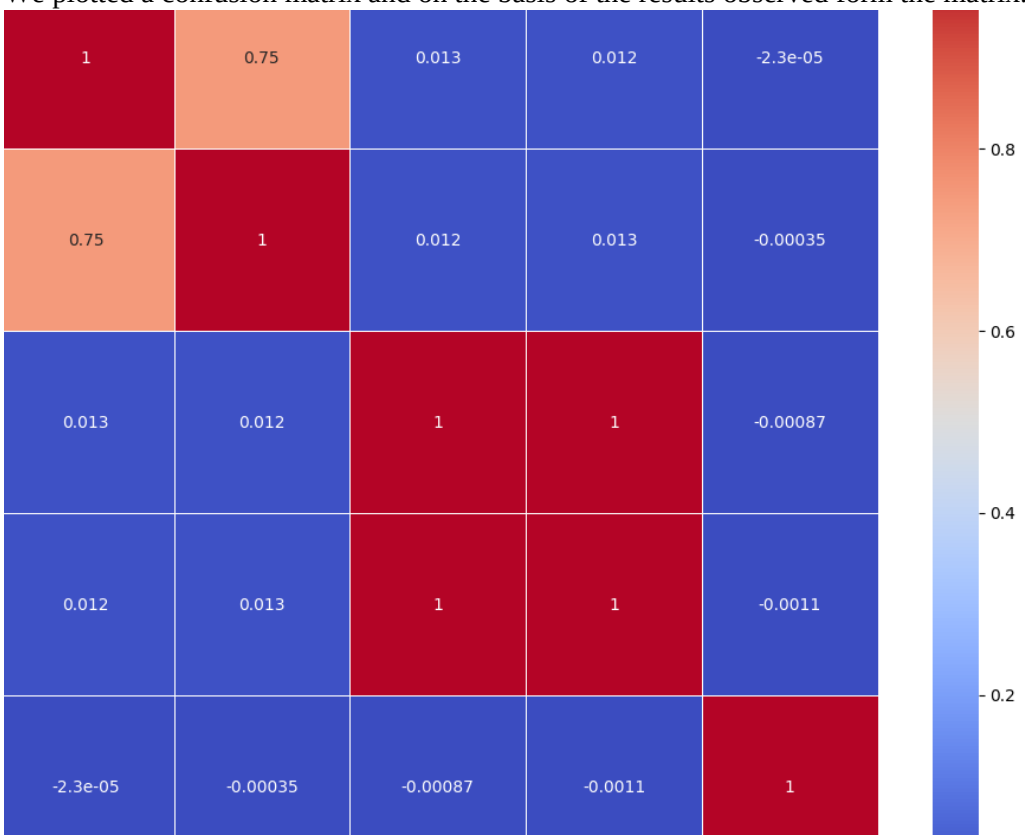
```
# Assuming df is your DataFrame
corr_matrix = df.corr()

# Set the size of the plot
plt.figure(figsize=(12, 10))

# Create a heatmap for all columns
sn.heatmap(corr_matrix, annot=True, cmap='coolwarm', linewidths=.5)

# Show the plot
plt.show()
```

We plotted a confusion matrix and on the basis of the results observed form the matrix.



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According to results seen from matrix we drop date.

```
#Dropping values
df.drop(['Date'], axis=1,inplace=True)
```

2) Null Values

We replaced the null values in Price with average of other values in price

```
average_price = df['Price($)'].mean()
df['Price($)'].fillna(average_price, inplace=True)
```

3) Trimming: Remove outliers based on statistical measures

```
# Assuming newdf is your DataFrame with the column "Annual Income"

# Convert the column to numeric type if it's not already numeric
newdf['Annual Income'] = pd.to_numeric(newdf['Annual Income'],
errors='coerce')

# Calculate z-scores for the column
z_scores = np.abs((newdf['Annual Income'] - newdf['Annual
Income'].mean()) / newdf['Annual Income'].std())

# Threshold for z-score based outlier removal (e.g., threshold = 3 for
3 standard deviations)
z_score_threshold = 3

# Mask to identify outliers based on z-scores
outlier_mask = z_scores > z_score_threshold

# Alternatively, you can use the interquartile range (IQR) method for
outlier removal
# Calculate the first quartile (Q1) and third quartile (Q3)
Q1 = newdf['Annual Income'].quantile(0.25)
Q3 = newdf['Annual Income'].quantile(0.75)
# Calculate the IQR
IQR = Q3 - Q1
```

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```
# Threshold for IQR-based outlier removal (e.g., threshold = 1.5)
IQR_threshold = 1.5
# Mask to identify outliers based on IQR
outlier_mask_iqr = (newdf['Annual Income'] < (Q1 - IQR_threshold *
IQR)) | (newdf['Annual Income'] > (Q3 + IQR_threshold * IQR))

# Remove outliers from the DataFrame using the mask
# For z-score method
newdf_no_outliers_z_score = newdf[~outlier_mask]
# For IQR method
newdf_no_outliers_iqr = newdf[~outlier_mask_iqr]

# Print the DataFrame without outliers for both methods
print("DataFrame without outliers (Z-score method):")
print(newdf_no_outliers_z_score)
print("\nDataFrame without outliers (IQR method):")
print(newdf_no_outliers_iqr)
```

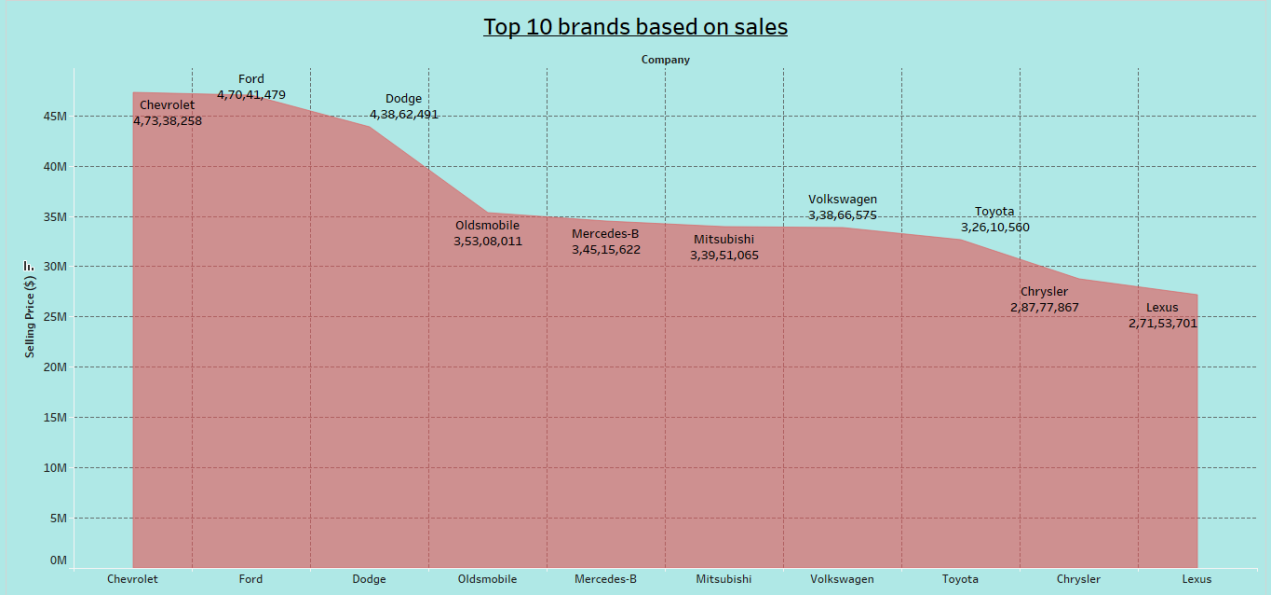
Here we showcases two distinct methods for detecting and removing outliers from a DataFrame containing cricket data, particularly focusing on the "Annual Income" column. The first method employs z-scores, which standardizes the values and identifies outliers based on a predefined threshold, typically set at 3 standard deviations from the mean. Alternatively, the IQR method utilizes quartiles to define a range within which most data points lie, flagging values outside this range as outliers. After detecting outliers using both methods, the code filters the DataFrame to exclude these anomalous data points. This systematic approach ensures the integrity of the dataset, facilitating more accurate analyses and insights into cricket performance metrics.

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Analysis and Visualization

1) Top 10 Brands

We found out the sales figures for all the brands in the dataset. For more clarity we applied the filter for that kept the top 10 Brands as Chevrolet, Ford, Dodge, Oldsmobile Mercedes-B, Mitsubishi, Volkswagen, Toyota, Chrysler and Lexus. As we can see Chevrolet is the largest company based on the sales figures.

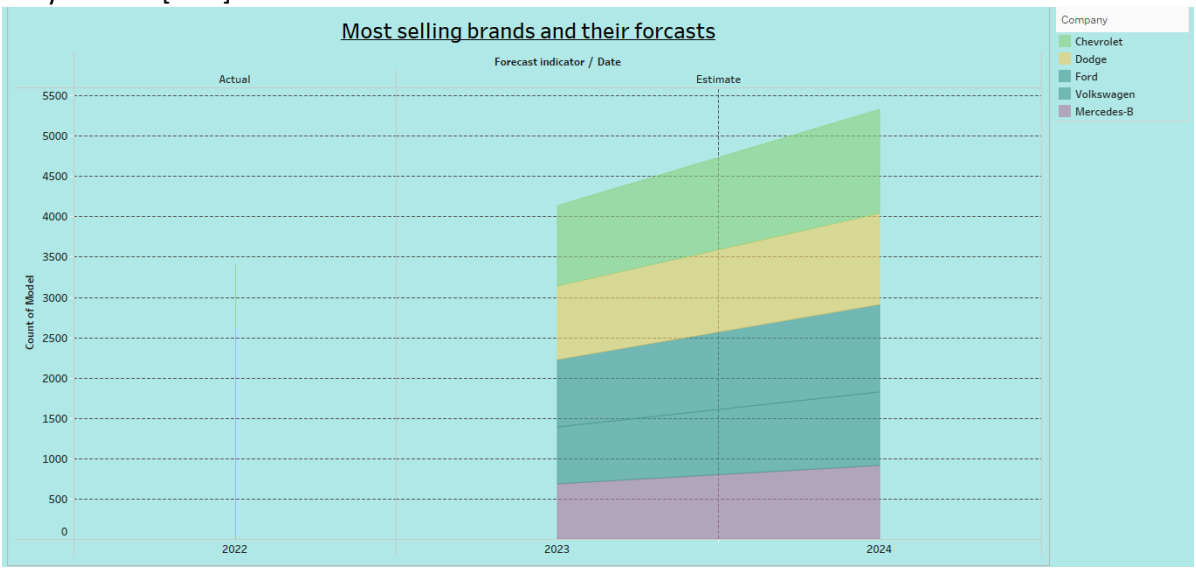


Here we are seeing the top 10 brands based on their sales figure of 2022(In $)

2) Top 5 Most Selling Brands

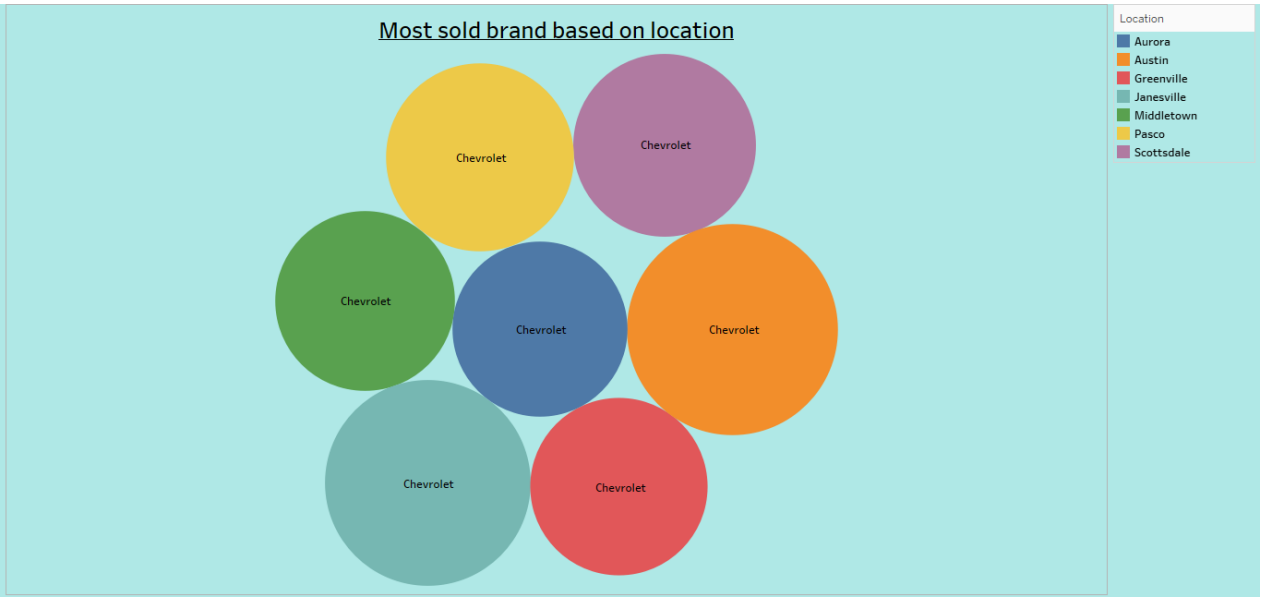
We found out the Brands forecast next year 2023. But for clarity we only took the top 5 companies and project the expected sales forecast. As we can again see Chevrolet is still on the top for the game.

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3) Most Sold Brands

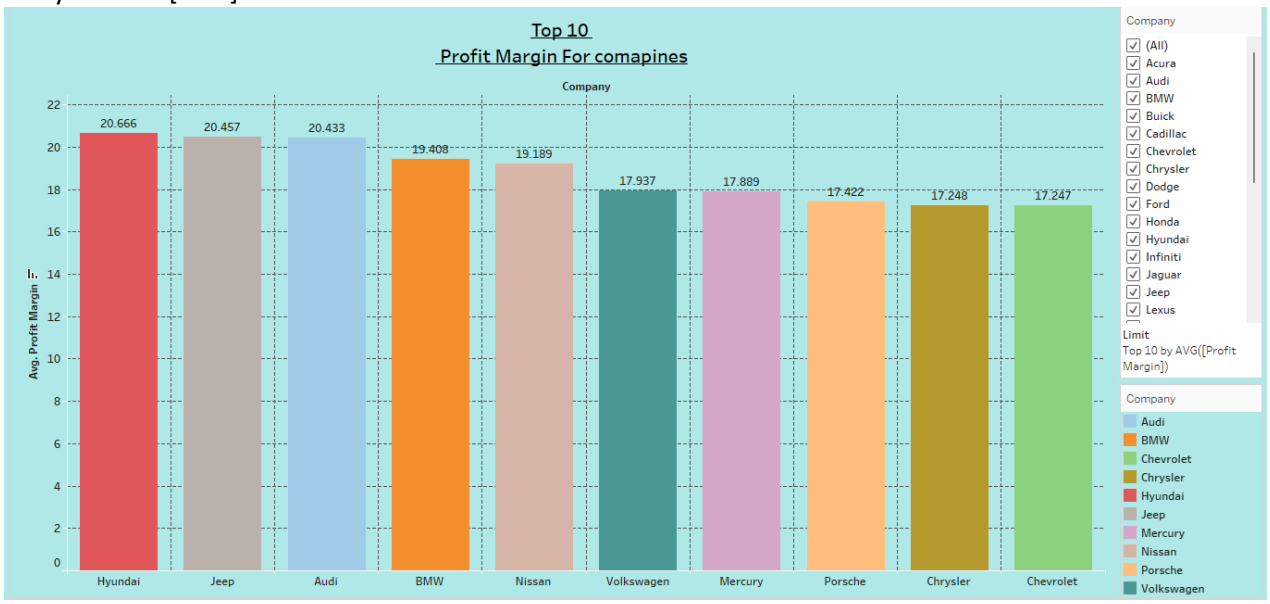
Now we saw Chevrolet is largest brand in America but is the largest in all the locations in America? So, we found out the most sold brand in each location and to everyone's surprise it was Chevrolet.



4) Profits Margin for the Companies

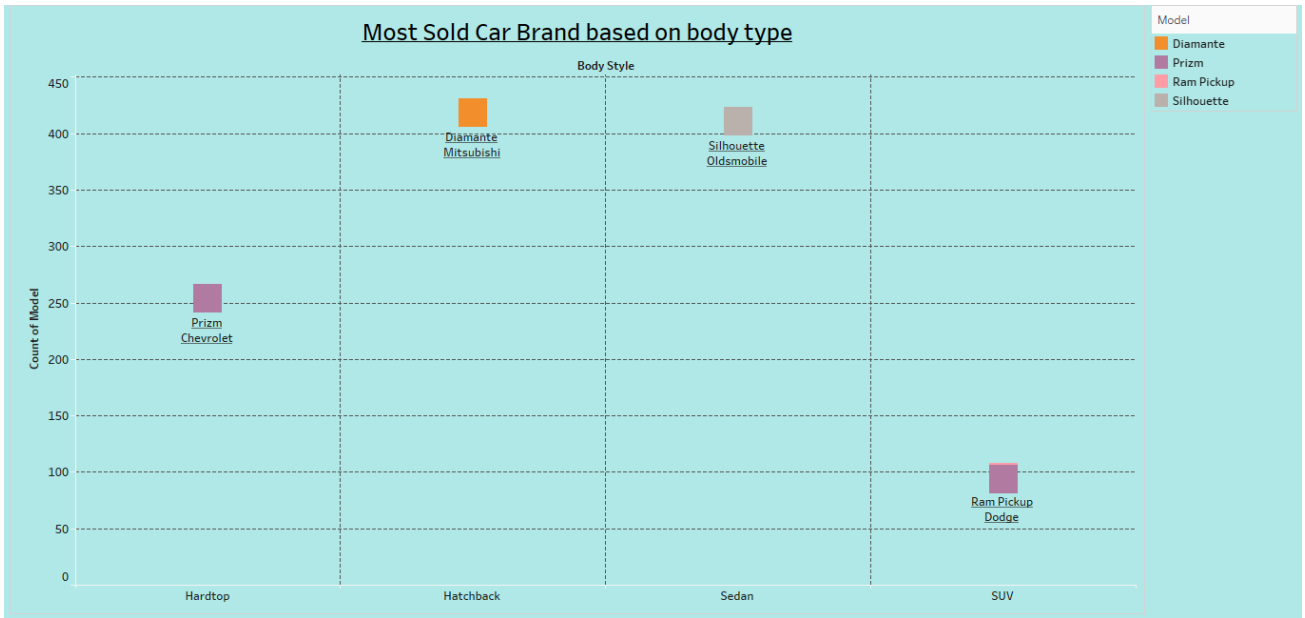
We saw Chevrolet is most famous brand in America but why? So, we thought their margins would be good, so every dealer is selling the brand but to our surprise is was on the 10th position in profit margins.

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5) Most Sold car for each body type

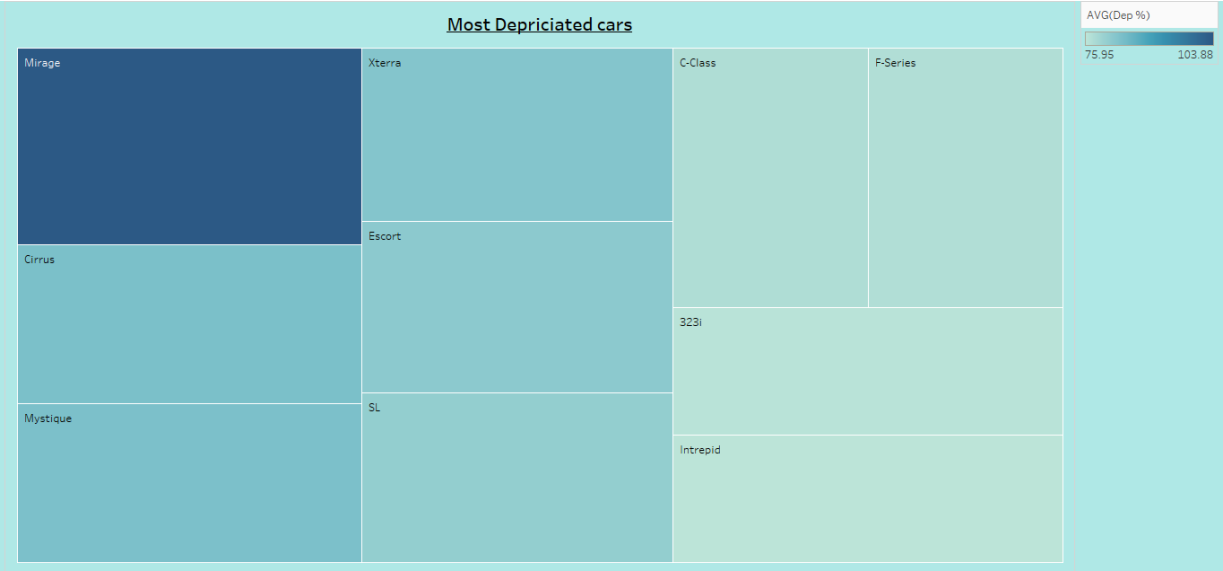
Now we wanted to see the most sold model for the everybody type category there is. So, we found out and to our surprise Chevrolet was not top in 3 of the body styles. So why is the most selling brand?



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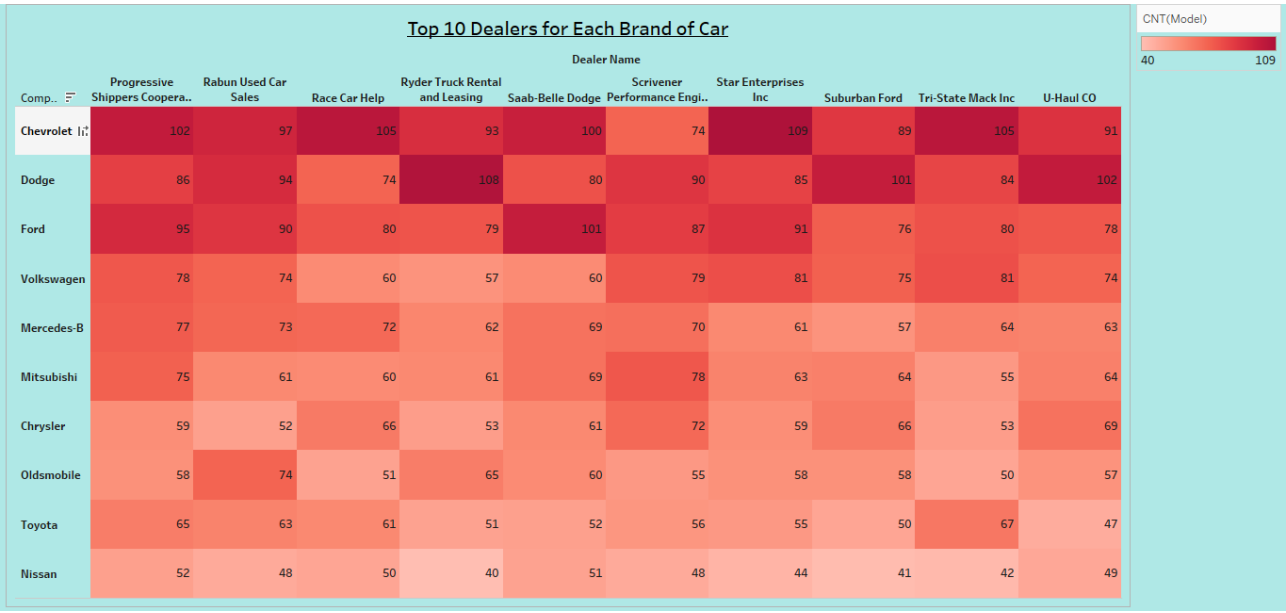
6) Most Depreciated cars

Now to answer our question why Chevrolet is so famous we found out the most depreciated cars of 2022 and it was not a single Chevrolet. This means that Chevrolet cars hold their values more and fetch higher second-hand price.



7) Top dealers for car brands

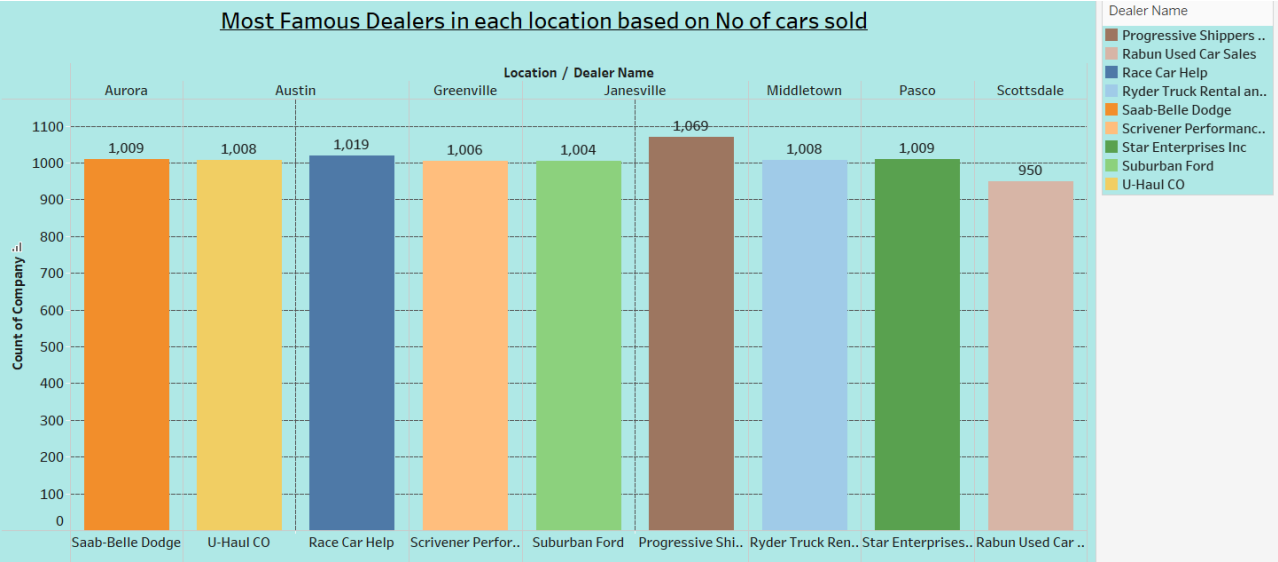
Since Chevrolet is so famous we wanted to see how the dealers stack up. Here we found out top 10 dealer for each brand and as we can see for the heat map Chevrolet has the dealers selling the cars the most.



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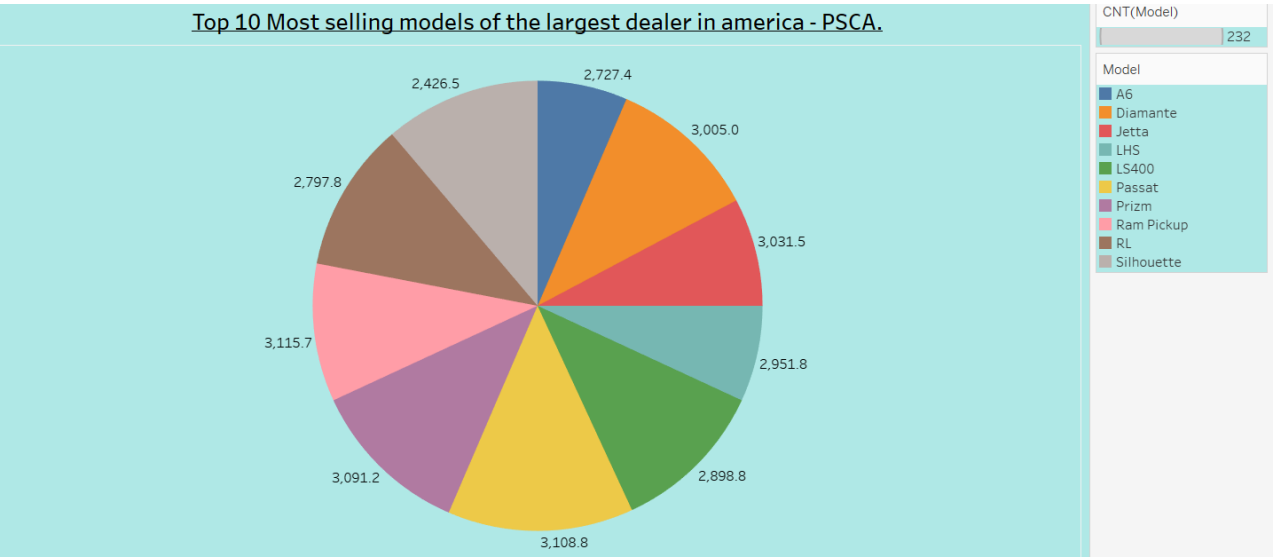
8) Most Famous Dealers in each location based on No of cars sold

Now that we have found out which company dominates the US car market we wanted to dig deeper into the market and find out who is largest dealer in US. We found the largest dealer to be Progressive Shipping Co. With 1069 units sold in 2022.

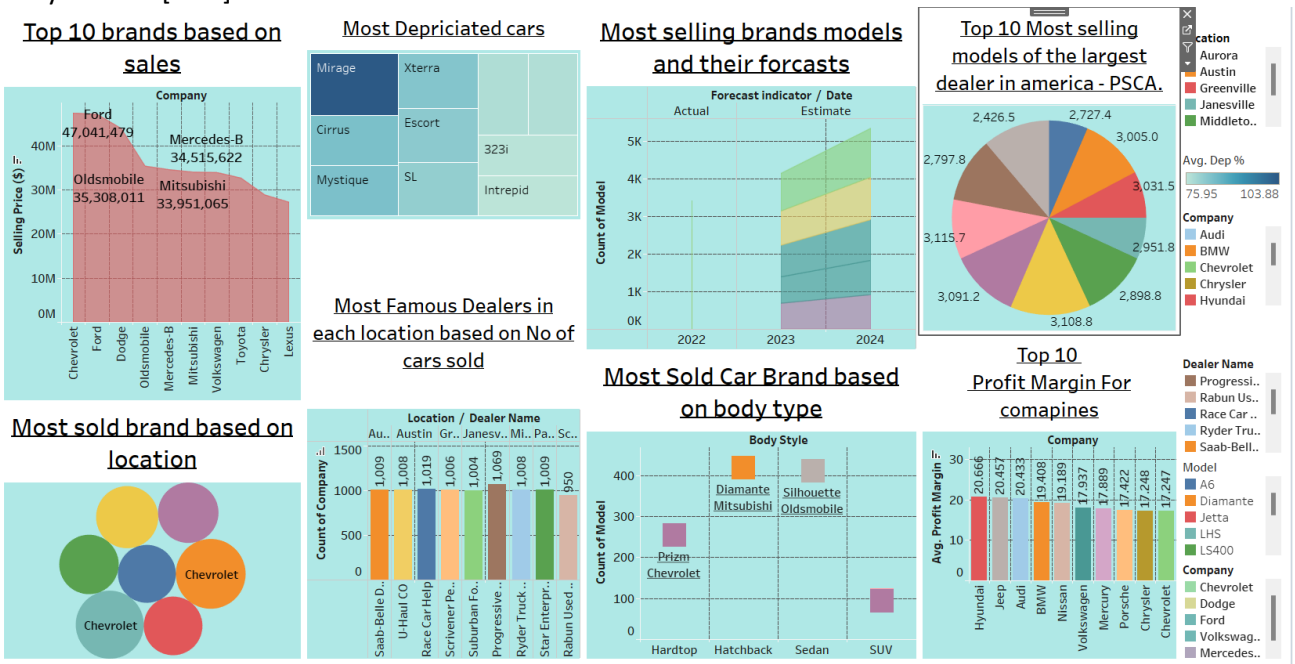


9) Top selling brands of the largest Dealer in US

Now that we have found the largest dealer and the largest company in US we wanted to see is the implications we found earlier were right. So, we found out the top selling models of the largest dealer in America and to no ones surprise they sell Chevrolet the most in 2022.



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Conclusion

In the fierce car market of US one brand stands out the most and this is Chevrolet. It sales for the year 2022 are through the roof, their cars hold up their value more and cost the customer less than the competition has no doubt made them the biggest car brand in the United States of America.